Víctor Manuel Chávez Pérez

Professor–Researcher · Applied Physics, Data Science & Science Education

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Google Scholar · ResearchGate · LinkedIn

National Researcher System · Candidate (SECIHTI) 2025–2029

PROFILE

I am a physicist from the University of Guadalajara and hold a PhD in Applied Physics from the University of Vigo. My research blends atmospheric and climate physics with large-scale data analysis: I study sudden stratospheric warmings (SSW), heat waves and their response under climate-change scenarios using models such as CMIP5 and WACCM.

Beyond basic research, I bring physics to real-world problems: I am developing a flood early-warning system for the Guadalajara Metropolitan Area that integrates weather radar, hydrological runoff modeling over high-resolution LiDAR terrain and remote sensing. My work also spans physical oceanography, remote sensing and applied optics.

I am also interested in how generative AI can transform science education: I design learning assistants and teaching materials that I evaluate with students themselves. Alongside this, I co-founded the Tlakakini Science & Technology Club, devoted to science outreach for children. I believe the best science is the kind that gets shared.

RESEARCH INTERESTS

Sudden stratospheric warmings · Stratospheric dynamics · General circulation models (CMIP5 / WACCM) · Climate change · Physical oceanography & remote sensing · Applied optics · Data science · High-performance computing · Science education & outreach · Generative AI in science education · Machine Learning applied to climate & atmospheric data · Student perception of AI-based learning tools

EDUCATION

2017 – 2024 **PhD in Applied Physics**
Universidade de Vigo · Spain
Thesis: Sensitivity of sudden stratospheric warmings to simulation and analysis conditions in models.

2011 – 2012 **MSc in Climate Science: Meteorology, Physical Oceanography & Climate Change**
Universidade de Vigo · Spain
Temporal and spatial variability of the Minho river plume (MODIS 250 m). Top honors.

2009 – 2011 **MSc in Hydrometeorology — Meteorology & Physical Oceanography**
Universidad de Guadalajara · Mexico
Seasonality and anomalies of oceanographic variables in the Gulf of Mexico via remote sensing.

2010 – 2011 **Diploma in Teaching Competencies**
Universidad de Guadalajara · Mexico
Pedagogical training for university teaching.

2000 – 2008 **BSc in Physics**
Universidad de Guadalajara · Mexico
Thesis: Ocean and coastal meteorology of Nayarit (San Blas, 1999–2002). Grade 95/100.

EXPERIENCE

2025 – present **Lecturer "B"**
Centro Universitario de Tlaquepaque · UdeG · Tlaquepaque, Mexico
Department of Basic Sciences. Chair of the Basic Sciences Academic Board.
Courses: Probability · Electromagnetism Lab · Discrete Mathematics · Numerical Methods · Calculus · Advanced Calculus

2025 – 2026 **Lecturer**
Centro Universitario Bonpland y Humboldt · Online
PhD in Aerospace Engineering: Research Methodology and Fluid Mechanics.
Courses: Research Methodology · Fluid Mechanics

2019 – present **Lecturer "B"**
Universidad Tecnológica de Jalisco · Guadalajara, Mexico
Mechatronics Division — Basic Sciences.
Courses: Engineering Physics · Engineering Mathematics I & II · Probability & Statistics · Integral Calculus · Applied Engineering Statistics · Thermodynamics

2019 – 2020 **Faculty Lecturer**
Tecmilenio · Campus Guadalajara · Guadalajara, Mexico
High school: Matter & Energy, Integral Calculus, Science & Technology.
Courses: Matter & Energy · Integral Calculus · Science & Technology

2019 **Physics Instructor**
UNITEC · Campus Guadalajara · Guadalajara, Mexico
Engineering programs: Statics.
Courses: Statics

2016 – 2024 **External Collaborating Researcher**
Universidade de Vigo · EPhysLab · Ourense, Spain
Doctoral research on sudden stratospheric warmings in CMIP5 and WACCM models; statistical analysis of large climate datasets.

2015 **Predoctoral mobility stay**
Universidad Pablo de Olavide · Seville, Spain
Short research stay at R&D centers; results presented at international conferences.

2013 – 2014 **Predoctoral research stay**
Universidad de Guadalajara · Guadalajara, Mexico
Academic research stay on climate-change analysis.

2012 – 2016 **Predoctoral Researcher (FPI Fellowship)**
Universidade de Vigo · EPhysLab · Ourense, Spain
Statistical analysis of large-volume data for climate-change research with general circulation models.

2012 – 2013 **Physics & Math Instructor**
Universidad de Especialidades · Guadalajara, Mexico
Mechanics, Introduction to Physics and Electromagnetism.
Courses: Mechanics · Intro to Physics · Electromagnetism

2010 – 2011 **Physics & Math Instructor**
Centro de Enseñanza Técnica Industrial (CETI Colomos) · Guadalajara, Mexico
Engineering programs.
Courses: Differential & Integral Calculus · Statics · Dynamics · Mathematics

2008 – 2011 **Professor–Researcher**
Instituto Tecnológico Superior de Zapopan · Zapopan, Mexico
Mathematics for engineering; parallel computing applied to oceanography.
Courses: Physics (Mechanics, E&M) · Differential & Integral Calculus · Differential Equations · Dynamics

PUBLICATIONS

2025 **Observación Astronómica en México: Pasando de lo Visible a lo Invisible**
Almaguer-Gómez C., Medina J. F., Chávez-Pérez V. M., et al. · *Óptica Pura y Aplicada 58(1):15* · DOI: 10.7149/OPA.58.1.51200

2025 **Influence of Satellite and Presatellite Periods on the Validation of Major Sudden Stratospheric Warmings in Historical CMIP5 Simulations**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 16(5)* · DOI: 10.3390/atmos16050628

2025 **Temporal Variability of Major Stratospheric Sudden Warmings in CMIP5 Climate Change Scenarios**
Chávez-Pérez V. M., et al. · *Climate 13(10)* · DOI: 10.3390/cli13100207

2023 **Estacionalidad y Anomalías del Nivel del Mar en el Caribe por Medio de Datos de Altimetría**
Palacios Hernández E., Chávez Pérez V. M., Sierra Romero L. L. · *Generis Publishing* · ISBN: 979-8-88676-642-4

2022 **Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Sudden Stratospheric Warmings**
Chávez-Pérez V. M., Añel J. A., et al. · *Atmosphere 13(4)* · DOI: 10.3390/atmos13040546

2018 **The climatology of dust events over the European continent using data of the BSC-DREAM8b model**
Mandija F., et al. · *Atmospheric Research 209:144–162* · DOI: 10.1016/j.atmosres.2018.03.006

In preparation

2026	Science Beyond the Classroom: Impact Assessment of a Monthly Science Club During National Non-School Days in Mexico Almaguer-Gómez C., Chávez-Pérez V. M., et al.
2026	Student Perception of a Generative AI-Based Calculus Learning Assistant: A Mixed-Methods Study Chávez-Pérez V. M., et al.
2026	Assessment of the lightning activity response to aerosol optical depth Mandija F., Chávez-Pérez V. M.

CONFERENCES & TALKS

2022

LXV Congreso Nacional de Física
2–7 oct 2022 · Zacatecas, Mexico
• Análisis de la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos en el Hemisferio Sur en los modelos de altura estratosférica del CMIP5
• Caracterización óptica de resina transparente de impresión 3D

2020

LXIII Congreso Nacional de Física
4–9 oct 2020 · Online
• Impacto en las significancias de las características básicas de los SSW al comparar modelos del CMIP5 con datos de reanálisis en periodos pre-satelital y pos-satelital
• Impacto en la frecuencia de ocurrencia de los calentamientos súbitos estratosféricos principales en distintos escenarios de cambio climático usando CMIP5

2018

American Geophysical Union (AGU) Fall Meeting 2018
10–14 dic 2018 · Washington, USA
• Impact of Increased Vertical Resolution in WACCM on the Climatology of Major Stratospheric Sudden Warmings

2018

Reunión Anual de la Unión Geofísica Mexicana
28 oct–2 nov 2018 · Puerto Vallarta, Mexico
• Cambios en la climatología de los calentamientos súbitos estratosféricos principales usando WACCM con alta resolución vertical
• Evaluación de la capacidad de reproducción de los calentamientos súbitos estratosféricos principales en el conjunto de datos históricos de CMIP5
• Calentamientos súbitos estratosféricos principales bajo escenarios de cambio climático con CMIP5

2017

European Geosciences Union (EGU) General Assembly 2017
23–28 abr 2017 · Vienna, Austria
• Estimation of the variability of dust aerosol optical depth over several European regions based on forecast model data

2016

LIX Congreso Nacional de Física (SMF)
2–7 oct 2016 · León, Mexico
• Teleconexión estratosférica y los calentamientos súbitos estratosféricos
• Patrones característicos en las variables estratosféricas de la Oscilación Cuasi-Bienal
• Anomalías estratosféricas antes y después de los calentamientos súbitos estratosféricos para las distintas fases de la Oscilación Cuasi-Bienal

2016

SPARC Workshop SHARP 2016 — Stratospheric Change and its Role for Climate Prediction
11–15 jul 2016 · Berlin, Germany
• Changes in the stratosphere before and after SSW events in the different phases of the Equatorial QBO
• Differences in the detection and classification of the Stratospheric Sudden Warming over the three reanalyses for the period 1979–2014

2015

European Geosciences Union (EGU) General Assembly
12–17 abr 2015 · Vienna, Austria
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2015

SPARC Regional Workshop — Role of the stratosphere in climate variability and prediction
12–13 ene 2015 · Granada, Spain
• Variability of the Brewer–Dobson circulation during Stratospheric Sudden Warmings

2010

Reunión Anual de la Unión Geofísica Mexicana
7–12 nov 2010 · Puerto Vallarta, Mexico
• Estacionalidad y anomalías de las variables de percepción remota oceanográficas en el Golfo de México
• Estacionalidad y anomalías de la superficie del nivel del mar en el Mar Caribe por medio de datos de altimetría
• Alternativa para el procesamiento de datos oceanográficos de gran cómputo mediante programación en paralelo con MatLab

2010 **LIII Congreso Nacional de Física**
25–29 oct 2010 · Boca del Río, Veracruz, Mexico
• Supercómputo aplicado a programación en paralelo para optimizar procesos en Astrofísica y Oceanografía Física

2009 **LII Congreso Nacional de Física (SMF)**
26–30 oct 2009 · Acapulco, Guerrero, Mexico
• Espectros de frecuencias de datos meteorológicos de Isla María Madre
• Comparación del viento costero y marino de la zona costera de San Blas, Nayarit (1999–2002)

ACTIVE RESEARCH PROJECTS

2026 – actual **Flood early-warning system for the Guadalajara Metropolitan Area — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Development of an urban flood early-warning system that combines remote sensing (IAM-UdeG weather radar), hydrological runoff modeling over high-resolution LiDAR terrain, and official flood inventories. The system classifies risk by sub-basin in real time and estimates water arrival time, aiming to calibrate thresholds through a retrospective analysis with historical Civil Protection/Fire Department records.

2026 - **Generative AI for lab practice reports (Google Gems) — Principal investigator**
Centro Universitario de Tlaquepaque de la Universidad de Guadalajara · Funding: No external funding
Implementation of Google Gems as a writing and feedback assistant for students to draft and self-assess their lab practice reports in physics courses.

2025 – actual **AI-generated educational comics for science teaching — Principal investigator**
Universidad de Guadalajara · Funding: No external funding
Development and evaluation of educational comics created with generative AI tools as a didactic resource for teaching physics, mathematics and science at basic and upper-secondary levels.

2025 – actual **Pedagogical Calculus Assistant with Google Gems: development and student perception — Principal investigator**
Universidad Tecnológica de Jalisco · Funding: Technological University of Jalisco
Design and implementation of a Calculus learning assistant based on Google Gems, with quantitative and qualitative study of students' perception, satisfaction and academic performance.

2025 – 2028 **Optical elements to enhance visual acuity via wavefront coding — Collaborator**
Universidad de Guadalajara · Funding: Universidad de Guadalajara
Design and validation of optical elements that improve visual acuity through wavefront coding.

2025 – actual **World Wide Lightning Location Network (WWLLN) — Collaborator · antenna lead**
University of Washington · Funding: University of Washington
Global lightning location network. In charge of the receiving antenna/node and real-time data delivery.

2024 – actual **Tlakakini Science & Technology Club — Coordinator**
CUCEI · Universidad de Guadalajara · Funding: University of Guadalajara · CUCEI
Science-outreach club for children, with monthly experiment and workshop sessions.

TECH DEVELOPMENTS

Flood early-warning system · Guadalajara Metro Area
Python · FastAPI · Leaflet · NumPy · pysheds · rasterio · GDAL · Docker
Real-time web platform that anticipates urban flooding in the Guadalajara Metropolitan Area. It fuses the IAM-UdeG weather radar (one scan per minute) with a hydrological runoff model built on 5 m LiDAR and Copernicus DEM terrain, plus the IMEPLAN official flood inventory (363 recurrent sites). It produces a per-zone risk light —calibrated against official water-depth records— and a continuous risk grid that routes rainfall downslope, with concentration time and a "water incoming" ETA. Includes live and historical-replay views. Deployed in production.

ACADEMIC SERVICE

Thesis supervision

2011 **Estacionalidad y anomalías del nivel del mar en el Caribe por medio de datos de altimetría**
Lilia Lizeth Sierra Romero · Bachelor's · Co-advisor · Universidad de Guadalajara

Journal peer review

2025 **IEEE Latin America Transactions**
Reviewer — Q2 journal, double-blind

2024–2025 **Int. J. of Environment and Pollution (Inderscience)**
Reviewer — Q4 journal, double-blind

2025 **Congreso de Ingeniería, Ciencia y Gestión Ambiental (AMICA)**
Scientific committee reviewer

Project evaluation

2025–2026 **SECIHTI**
External evaluator — 2026 National Call, Basic & Frontier Science (3 proposals)

SCIENCE OUTREACH

2024 – **Tlakakini Science & Technology Club**
Co-founder. Monthly outreach sessions for children on national non-school days.

2009 – **Physics conferences (AGU, EGU, SMF)**
Talks at national and international conferences on climate and the stratosphere.

2005 – **Training science communicators**
Optics and electromagnetism workshops; National Outreach Meeting.

2009 – **Environment & Sustainability Thematic Network**
Member of the CONACYT environment and sustainability network.

2011 – **Environmental Physics Laboratory (EPhysLab)**
External collaborating researcher with the University of Vigo group.

GRANTS & HONORS

2025–2029 **National Researcher System · Candidate**
SECIHTI · Mexico

2012–2016 **FPI Grant · Predoctoral Researcher**
Ministerio de Educación · Spain

2011–2012 **Climate Science Master's Scholarship**
Campus do Mar · Spain

2011–2012 **«Ourense Exterior» mobility grant**
Diputación de Ourense · Spain

2009–2011 **Hydrometeorology Master's Scholarship**
CONACYT · Mexico

2009 **Admission to the State Research System**
COECYTJAL · Jalisco, Mexico

CONTINUING EDUCATION

2025 **Diploma** — Estrategias tecnopedagógicas para el desarrollo de habilidades del siglo XXI · Universidad de Guadalajara

2025 **Course** — Inteligencia Artificial para la docencia · Universidad de Guadalajara

2025 **Course** — Inteligencia Artificial Generativa para Docentes · Universidad Anáhuac

2025 **Course** — IA Generativa en el aula · UNAM

2025 **Course** — Python for Data Science, AI & Development · IBM

2025 **Certification** — What is Data Science? · IBM

2025 **Course** — Cálculo Diferencial para Data Science e IA · Platzi

2025 **Course** — Estadística y Probabilidad · Platzi

2025 **Course** — Fundamentos de Ingeniería de Software · Platzi

2025 **Course** — Introducción a la Terminal y Línea de Comandos · Platzi

2014 **Course** — OOP in Atmospheric and Oceanic Model Development · Universidade de Vigo

2011 **Diploma** — Competencias Docentes: Inducción al Bachillerato General · Universidad de Guadalajara

2005 **Workshop** — Formación de Divulgadores de la Ciencia · Universidad de Guadalajara

TOOLBOX

Python · Fortran · MatLab · IDL · NCL · LaTeX · COMSOL · LabVIEW · MPI · Linux · Moodle · Pandas · Scikit-learn · Jupyter